Lösung zu Kapitel 10, mongoDB-Aufgaben

(3) Erstellen Sie nun ein erstes **weibliches** Wesen mit Name **Aurora** und Gewicht **450** in der Collection `unicorns`

Lösungsvorschlag:

```
db.unicorns.insert({name: 'Aurora', gender: 'f', weight: 450})

>>> WriteResult({ "nInserted" : 1 })
```

(4) Erstellen Sie eine erste Abfrage zur Überprüfung des erstellten Dokuments in der Collection `unicorns`

Lösungsvorschlag:

```
db.unicorns.find()

>>> { "_id" : ObjectId("58ef0f139f1556b5cea43115"), "name" :
"Aurora", "gender": "f", "weight" : 450 }
```

(5) Fügen Sie nun nach belieben weitere Fabelwesen ein und variieren Sie dabei auch die Anzahl und Typen der verschiedenen Felder im Dokument, z.B.

Lösungsvorschlag:

```
db.unicorns.insert({name: 'Arris', gender: 'm', loves:
['apple','grape'], weight: 610})

>>> WriteResult({ "nInserted" : 1 })

db.unicorns.insert({name: 'NewDat', gender: 'm', dob: new
Date(2013,04,14,01,02), weight: 25})

>>> WriteResult({ "nInserted" : 1 })
```

(6) Lassen Sie sich nun nochmals alle bis jetzt eingefügten Fabelwesen anzeigen.

Lösungsvorschlag:

```
db.unicorns.find()

>>> { "_id" : ObjectId("516bef4544791acce5ec75b3"), "name" :
"Aurora", "gender": "f", "weight" : 450 }
>>> { "_id" : ObjectId("516beff744791acce5ec75b5"), "name" : "Arris",
"gender": "m", "loves" : [ "apple", "grape" ], "weight" : 610 }
>>> { "_id" : ObjectId("516bf04544791acce5ec75b6"), "name" :
"NewDat", "gender": "m", "dob" : ISODate("2013-05-13T23:02:00Z"),
"weight" : 25 }
```

(7) Löschen Sie nun alle Dokumente der Collection `unicorns`

Lösungsvorschlag:

```
db.unicorns.remove({})
```

```
>>> WriteResult({ "nRemoved" : 4 })
```

(8) Daten aus einer Datei einfügen

```
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4000"), name:
'Horny', dob: new Date(1992,2,13,7,47), loves: ['carrot','papaya'],
weight: 600, gender: 'm', vampires: 63});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4001"), name:
'Aurora', dob: new Date(1991, 0, 24, 13, 0), loves: ['carrot',
'grape'], weight: 450, gender: 'f', vampires: 43});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4002"), name:
'Unicrom', dob: new Date(1973, 1, 9, 22, 10), loves: ['energon',
'redbull'], weight: 984, gender: 'm', vampires: 182});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4003"), name:
'Roooooodles', dob: new Date(1979, 7, 18, 18, 44), loves: ['apple'],
weight: 575, gender: 'm', vampires: 99});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4004"), name:
'Solnara', dob: new Date(1985, 6, 4, 2, 1), loves:['apple', 'carrot',
'chocolate'], weight:550, gender:'f', vampires:80});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4005"), name:
'Ayna', dob: new Date(1998, 2, 7, 8, 30), loves: ['strawberry',
'lemon'], weight: 733, gender: 'f', vampires: 40});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4006"), name:
'Kenny', dob: new Date(1997, 6, 1, 10, 42), loves: ['grape',
'lemon'], weight: 690,  gender: 'm', vampires: 39});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4007"), name:
'Raleigh', dob: new Date(2005, 4, 3, 0, 57), loves: ['apple',
'sugar'], weight: 421, gender: 'm', vampires: 2});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4008"), name:
'Leia', dob: new Date(2001, 9, 8, 14, 53), loves: ['apple',
'watermelon'], weight: 601, gender: 'f', vampires: 33});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac4009"), name:
'Pilot', dob: new Date(1997, 2, 1, 5, 3), loves: ['apple',
'watermelon'], weight: 650, gender: 'm', vampires: 54});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac400a"), name:
'Nimue', dob: new Date(1999, 11, 20, 16, 15), loves: ['grape',
'carrot'], weight: 540, gender: 'f'});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac400b"), name:
'Dunx', dob: new Date(1976, 6, 18, 18, 18), loves: ['grape',
'watermelon'], weight: 704, gender: 'm', vampires: 165});
db.unicorns.insert({_id: ObjectId("5cc6e5d94249f49a59ac400c"), name:
'Calina', dob: new Date(1987, 8, 10, 5, 56), loves: ['chocolate',
'watermelon'], weight: 340, gender: 'f', vampires: 0});

db.unicorns.countDocuments()
```

(9) Lassen Sie sich nun die Wesen mit den jeweils folgenden Eigenschaften anzeigen:

a) mit Namen Aurora
Lösungsvorschlag:

```
db.unicorns.find({name:'Aurora'})

>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4001"), "name" :
"Aurora", "dob" : ISODate("1991-01-24T13:00:00Z"), "loves" : [
"carrot", "grape" ], "weight" : 450, "gender" : "f", "vampires" : 43
}
```

b) alle Weibchen, welche Äpfel mögen
Lösungsvorschlag:

```
db.unicorns.find({gender:'f',loves:'apple'})

>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4004"), "name" :
"Solnara", "dob" : ISODate("1985-07-04T02:01:00Z"), "loves" : [
"apple", "carrot", "chocolate"], "weight" : 550, "gender" : "f",
"vampires" : 80 }
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4008"), "name" : "Leia",
"dob" : ISODate("2001-10-08T14:53:00Z"), "loves" : [ "apple",
"watermelon" ], "weight" : 601, "gender" : "f", "vampires" : 33 }
```

c) alle Weibchen, welche Äpfel und Karotte mögen
Lösungsvorschlag: (Zur Erinnerung: Auf manchen Cloud-Plattformen sind hier und nachfolgend alle $ durch \$ zu ersetzen.)

```
db.unicorns.find({gender:'f',loves:{ $all:['apple','carrot']}})
// oder
db.unicorns.find({gender:'f',$and:[{loves:'apple'},{loves:'carrot'}]}
)
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4004"), "name" :
"Solnara", "dob" : ISODate("1985-07-04T02:01:00Z"), "loves" : [
"apple", "carrot", "chocolate"], "weight" : 550, "gender" : "f",
"vampires" : 80 }
```

d) alle Weibchen, welche Äpfel oder Karotte mögen.

Lösungsvorschlag:

```
db.unicorns.find({gender:'f',loves:{ $in:['apple','carrot']}})
// oder
db.unicorns.find({gender:'f',$or:[{loves:'apple'},
{loves:'carrot'}]})

>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4001"), "name" :
"Aurora", "dob" : ISODate("1991-01-24T13:00:00Z"), "loves" : [
"carrot", "grape" ], "weight" : 450, "gender" : "f", "vampires" : 43
}
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4004"), "name" :
"Solnara", "dob" : ISODate("1985-07-04T02:01:00Z"), "loves" : [
"apple", "carrot", "chocolate"], "weight" : 550, "gender" : "f",
"vampires" : 80 }
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4008"), "name" : "Leia",
"dob" : ISODate("2001-10-08T14:53:00Z"), "loves" : [ "apple",
"watermelon" ], "weight" : 601, "gender" : "f", "vampires" : 33 }
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac400a"), "name" : "Nimue",
"dob" : ISODate("1999-12-20T16:15:00Z"), "loves" : [ "grape",
"carrot" ], "weight" : 540, "gender" : "f" }
```

e) alle Wesen, welche bis jetzt noch *keine* Vampire gesehen haben.

Lösungsvorschlag:

Dies ist ein typischer Fall einer schemalosen Dokumentendatenbank. Wir müssen auch Dokumente suchen, welche noch keinen Eintrag `vampires` beinhalten.

```
db.unicorns.find({$or:[{vampires: {$exists:false}},{vampires:
{$lte:0}}]})
// oder
db.unicorns.find({\$or:[{vampires: {\$exists:false}},{vampires: 0}]})

>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac400a"), "name" : "Nimue",
"dob" : ISODate("1999-12-20T15:15:00Z"), "loves" : [ "grape",
"carrot" ], "weight" : 540, "gender" : "f" }
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac400c"), "name" :
"Calina", "dob" : ISODate("1987-09-10T03:56:00Z"), "loves" : [
"chocolate", "watermelon" ], "weight" : 340, "gender" : "f",
"vampires" : 0 }
```

f) alle Männchen, welche ihr Gewicht zwischen 600 und 900 haben.

Lösungsvorschlag.

```
db.unicorns.find({gender:'m',$and:[{weight:{$gte:600}},{weight:{$lte:
900}}]})

>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4000"), "name" : "Horny",
"dob" : ISODate("1992-03-13T06:47:00Z"), "loves" : [ "carrot",
"papaya" ], "weight" : 600, "gender" : "m", "vampires" : 63 }
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4006"), "name" : "Kenny",
"dob" : ISODate("1997-07-01T08:42:00Z"), "loves" : [ "grape", "lemon"
], "weight" : 690, "gender" : "m", "vampires" : 39 }
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac4009"), "name" : "Pilot",
"dob" : ISODate("1997-03-01T04:03:00Z"), "loves" : [ "apple",
"watermelon" ], "weight" : 650, "gender" : "m", "vampires" : 54 }
>>> { "_id" : ObjectId("5cc6e5d94249f49a59ac400b"), "name" : "Dunx",
"dob" : ISODate("1976-07-18T17:18:00Z"), "loves" : [ "grape",
"watermelon" ], "weight" : 704, "gender" : "m", "vampires" : 165 }
```

Dokumentation der Query Operatoren

Query Operatoren

```
$regex Match by any PCRE-compliant regular expression string (or just use
the // delimiters as shown earlier)
$ne Not equal to
$lt Less than
$lte Less than or equal to
$gt Greater than
$gte Greater than or equal to
$exists Check for the existence of a field
$all Match all elements in an array
$in Match any elements in an array
$nin Does not match any elements in an array
$elemMatch Match all fields in an array of nested documents
$or or
$nor Not or
$size Match array of given size
$mod Modulus
$type Match if field is a given datatype
$not Negate the given operator check
```